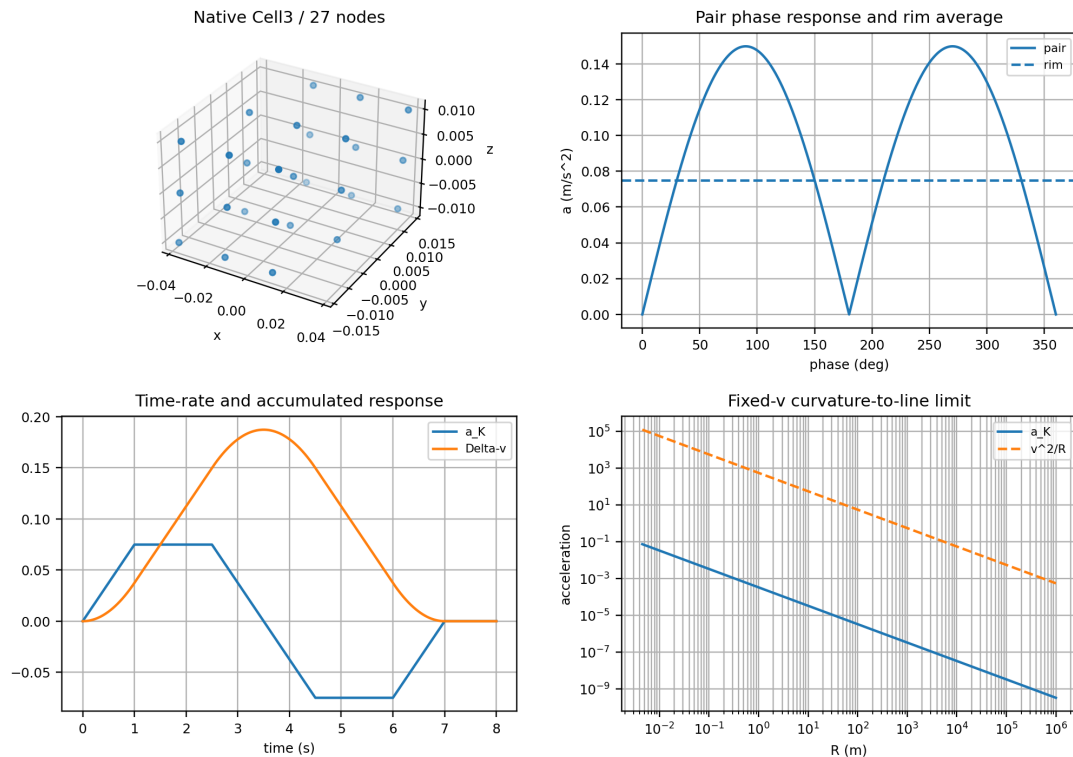


K-FIELD

Time-Force Dependency

Angular phase rate, antipodal reduction, curvature-to-line limit, and accumulated free-space motion

K-Field time-force dependency: exact computed structures



Title figure. Every panel is generated from the authoritative Cell3 geometry, calibrated Earth-CMB field inputs, the defined surgical-stainless tube, and exact computations of phase, force, acceleration, and the large-radius limit.

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This paper is a standalone educational and archival reference. It preserves the complete time-force analysis, develops its mathematical basis from the native Cell3 antipodal geometry, and separates phase-domain invariants from their time-domain delivery. All numerical figures and tables are generated from the attached source ledgers and exact computations on those values.

Abstract

The integrated K-Field force of a rotating tubular mass is governed by signed angular phase rate. For a fixed spin axis, the acceleration law is $a_K(t) = (\kappa_F/2) \Omega(t) \times g$, where $g = G_{\text{phase}} V_E$ and $\Omega = d\phi/dt$. This makes time unavoidable at the constitutive level: a stationary geometry produces no force, reversing temporal phase progression reverses the force, and the force magnitude measures phase traversed per unit time. The phase-domain velocity increment $d v_K / d\phi = (\kappa_F/2) \hat{n} \times g$ is invariant, while physical acceleration is that invariant multiplied by phase rate. A discrete antipodal pair displays a 2Ω ripple around the steady mean; a complete uniform rim is the simultaneous all-phase integral that removes the ripple and preserves the mean. In the controlled remote-center limit R to infinity with branch speed v fixed, $\Omega = v/R$ tends to zero and both ordinary centripetal acceleration and K-Field rotational acceleration vanish. The resulting law is therefore a curvature-weighted momentum or turning-rate response, not a response to uniform translation alone.

Central result

$$\begin{aligned} a_K(t) &= (\kappa_F / 2) \Omega(t) \times g \\ \Omega(t) &= d \phi(t) / dt \\ d v_K / d \phi &= (\kappa_F / 2) \hat{n} \times g \\ \Delta v_{\text{per rev}} &= \pi \kappa_F \hat{n} \times g \end{aligned}$$

The geometry fixes the direction and the velocity increment per signed angular phase. Time fixes the rate at which that increment is delivered.

Contents

Section	Role
1. Scope and reference configuration	Definitions and source lock
2. Native Cell3 geometry	Base cell, face planes, and 27 nodes
3. Earth-CMB field and orientation lock	ICRS vectors and optimal force direction
4. Defined tubular rotor	Mass, inertia, spin energy, and angular momentum
5. Antipodal reduction	Exact two-point surrogate and rim completion
6. Turning rate as the primitive variable	From phase to time-domain force
7. Phase-domain invariant	Velocity increment per signed angular phase
8. Steady operation versus time independence	Why a constant force remains rate-dependent
9. Explicit antipodal-pair time response	Mean term and 2 Omega ripple
10. Circumferential integration	Rim cancellation and steady mean
11. Time reversal and signed phase	Start, stop, and reverse laws
12. Large-radius curvature-to-line limit	Fixed-v remote-center asymptotics
13. Accumulated free-space motion	Velocity and displacement laws
14. Spin-state and energy relations	Inertia, energy, momentum, and acceleration
15. Dimensional and structural interpretation	Turning-rate and vorticity forms
16. Experimental observables	Direct tests and discriminating signatures
17. Conclusions	Complete hierarchy
Appendix A. Matrix and vector ledger	Reference data
Appendix B. All 27 Cell3 nodes	Complete address table
Appendix C. Formula ledger	Machine reconstruction equations
Appendix D. Source and reproducibility record	Files, hashes, and generation record

1. Scope and reference configuration

The analysis concerns one uniform hollow tube rotating about its own longitudinal axis. The tube is free in space unless a mounting constraint is explicitly introduced. Its spin axis is oriented to maximize the K-Field acceleration while preserving exact orthogonality to the Earth CMB traversal direction.

The reference tube has outer diameter 10 mm, wall thickness 1 mm, inner diameter 8 mm, length 100 mm, and working material density 8000 kg/m^3 . It rotates at 50,000 rpm. The resulting force is derived from the continuous-rim integral and is then reduced to an inertia-equivalent antipodal pair.

Source role	File	SHA-256
free_space_json	K-Field_tubular_rotor_free_space_forces_1784904809.json	205a825782a75d3ff51a70cf926b3220e687e8bcd11afcce9952d68f7d66f264
free_space_pdf	K-Field_tubular_rotor_free_space_forces_1784904809.pdf	bcef94c8ba6a939970dc015d448a728c0ffda46f9c62e008e17d064a2b9f53e5
tube_force_json	K-Field_tubular_rotor_force_analysis_1784897756.json	898e9433dc4930ef364392811bd324a629c1a8b7075d239b180aa05cc977e665
tube_force_pdf	K-Field_tubular_rotor_force_analysis_1784897756.pdf	0f59d2a605c896f918fdf9df809afa43a2deab8af228d8e14025d5c4b1ba2680
sphere_consequences_json	K-Field_Consequences_of_the_Native_Cell3_Antipodal_Sphere_Theorem_1784494389(2).json	6e7004c122f1c9e96424e7f55b931236e37e08be36ca3239718b11c1d6ffed07
sphere_theorem_json	K-Native_Cell3_antipodal_sphere_theorem1784492216(2).json	2c6331de64d582f58b287f8bda04ca2cf90e70e7eae1c1d463b5f1ec8cfa5005
remote_center_pdf	K-Field_mirror_rotation_about_remote_projected_center_1784227730(2).pdf	cee085f383dff3b26e3107b4f387459a67b46b524ab90dc50f9bcdbe783c5a7a

The attached native theorem establishes the projective antipodal structure used here: an occupied origin-centered Cell3 sphere contains exactly one nonzero lattice point and its antipode, and an origin-centered circle through that point contains the same pair. The continuous rim is consequently interpreted as the phase completion of the primitive antipodal channel rather than as a set of unrelated equal-radius lattice centers.

2. Native Cell3 geometry

Cell3 is represented in an orthonormal study frame by the basis matrix $B=[a \ b \ c]$. Every centered Lucas address $L=(L1,L2,L3)$, with each coordinate in $\{-1,0,+1\}$, maps to $p_L=B \ L/2$. This gives 27 nodes: eight corners, twelve edge centers, six face centers, and one body center.

Quantity	Value	Dimension
a	0.0207970587824329	native
b	0.0364942051307272	native
c	0.0493085658998594	native
A_ab	0.000754721789672624	native^2
A_ac	0.00101125575768807	native^2
A_bc	0.00138578443173329	native^2
Volume	2.84206693780965e-05	native^3

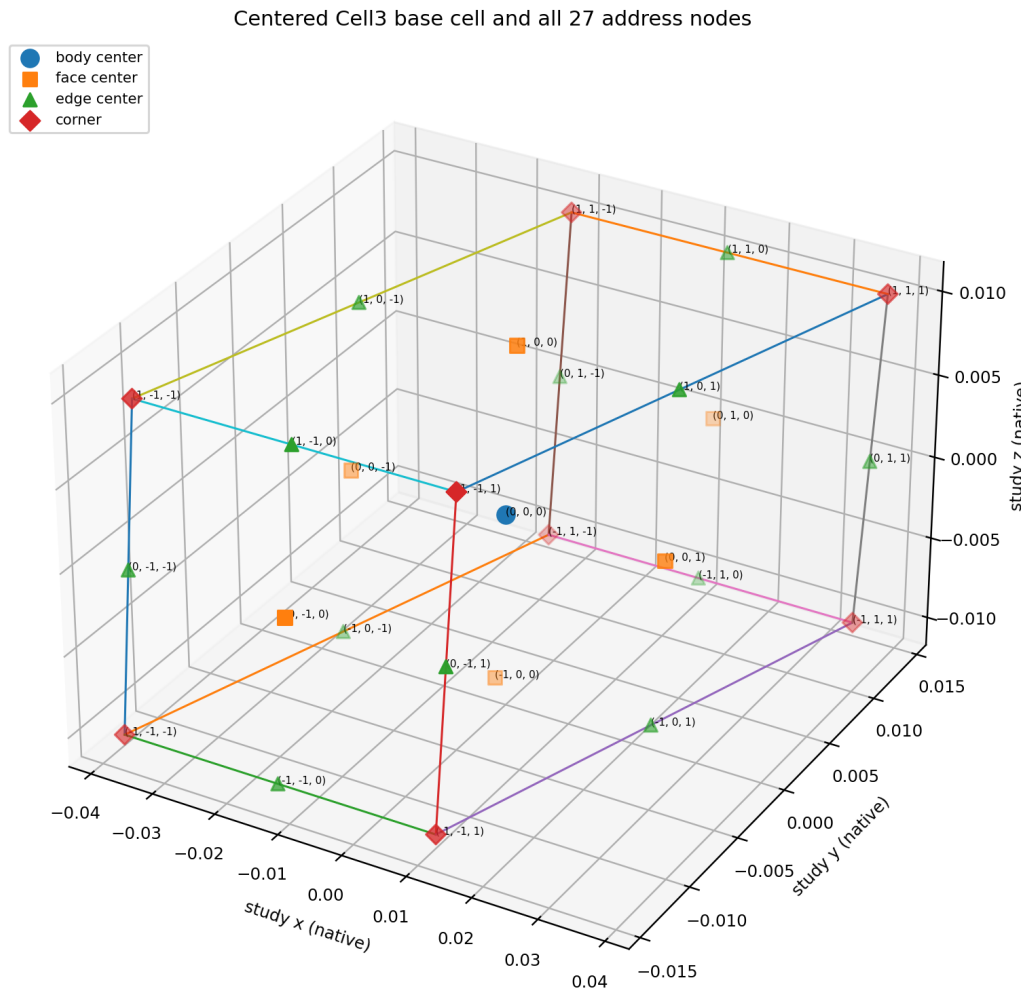


Figure 1. Dimensionally correct centered Cell3 in the orthonormal study frame. All 27 supplied address nodes are plotted and labeled by $L=(L1,L2,L3)$. The cell edges are generated directly from the supplied basis vectors.

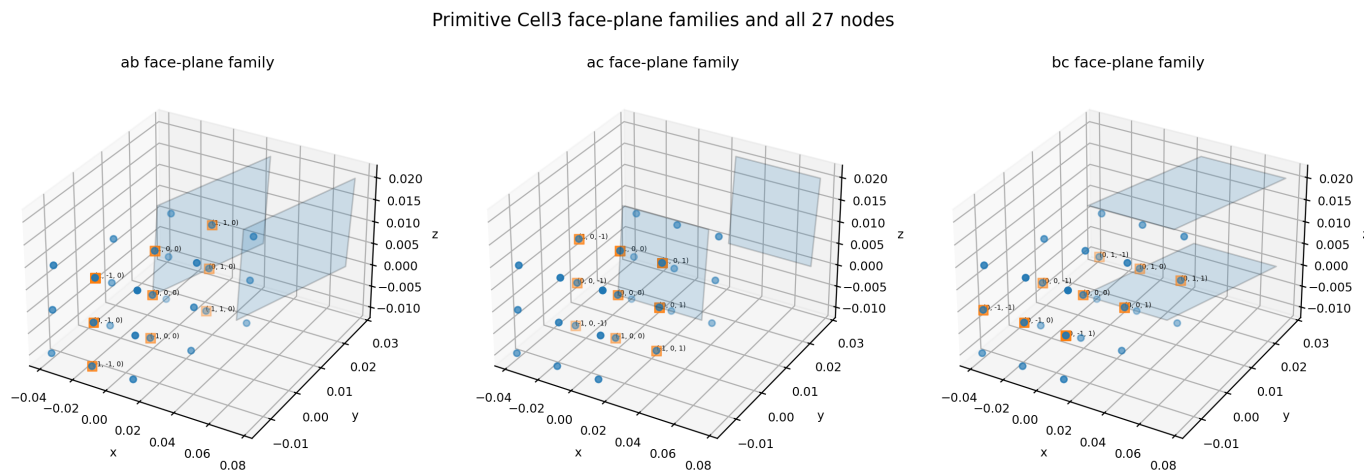


Figure 2. Primitive ab, ac, and bc face-plane families. Each panel shows both boundary planes, all 27 Cell3 nodes, and the nodes lying on the corresponding plane family.

The Cell3 sphere theorem defines each nonzero exact radial channel projectively: the two members are p and $-p$. The theorem therefore supplies the discrete antipodal primitive. Rotation of a physical continuous mass around such an axis is a continuous phase filling of that primitive pair, not an enlargement of the exact center-lattice multiplicity.

3. Earth-CMB field and orientation lock

The physical response vector is $\mathbf{g} = \mathbf{G_phase} \mathbf{V_E}$. The fixed ICRS/J2000 coordinate frame is used throughout. The Earth CMB unit vector $\mathbf{c_hat}$, the response vector $\mathbf{g}$, the optimal tube axis $\mathbf{n_opt}$, and the force direction $\mathbf{f_hat}$ form a locked vector construction.

```
g = G_phase V_E
F_tube = (kappa_F M omega / 2) n_hat x g
a_tube = F_tube / M = (kappa_F omega / 2) n_hat x g
```

Vector	ICRS Cartesian	Direction
Earth CMB $\mathbf{c_hat}$	(np.float64(-0.9707708558359351), np.float64(0.20734800911098614), np.float64(-0.12087492948199192))	RA 167.943300 deg; Dec -6.942600 deg
$\mathbf{g_hat}$	(np.float64(-0.9456361573618158), np.float64(-0.15460629330993783), np.float64(0.2861278594596136))	RA 189.285391 deg; Dec 16.626278 deg
optimal $\mathbf{n_hat}$	(np.float64(-0.3158650364073233), np.float64(0.6461626830235584), np.float64(-0.6947683540887543))	RA 116.050867 deg; Dec -44.008761 deg
force / acceleration $\mathbf{f_hat}$	(np.float64(0.07746958542150294), np.float64(0.7473758633624434), np.float64(0.6598695190701438))	RA 84.082114 deg; Dec 41.289922 deg
Quantity	Value	Dimension
$ \mathbf{V_E} $	369812.502	m/s
$ \mathbf{g} $	930825.21727	m/s
κ_F	3.0740508997020345e-11	calibrated coupling
$q_g = \kappa_F \mathbf{g} /2$	1.4307020483077610e-05	N/[kg (rad/s)]

Exact ICRS vector lock for maximum CMB-orthogonal acceleration

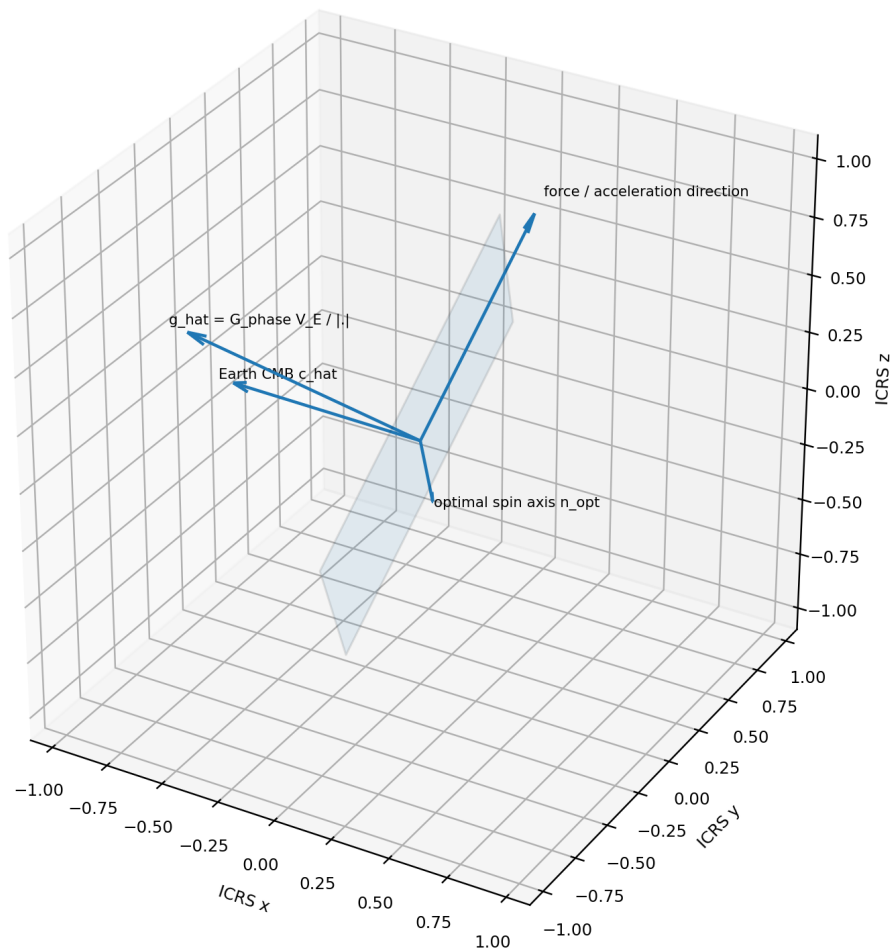


Figure 3. Exact ICRS vector geometry. The optimal tube axis lies in the plane perpendicular to $\mathbf{g}$, while the resulting force is parallel to $\mathbf{n}_{opt}$ cross $\mathbf{g}$ and perpendicular to the Earth CMB vector to numerical precision.

4. Defined tubular rotor

The tube mass is determined by the annular metal area and length. Its axial moment of inertia controls the torque and energy required to establish the specified spin; it does not enter the free acceleration after division by mass.

$$\begin{aligned}
 M &= \rho \pi (r_o^2 - r_i^2) L \\
 I_{spin} &= (M/2) (r_o^2 + r_i^2) \\
 H_{spin} &= I_{spin} \omega \\
 E_{spin} &= (1/2) I_{spin} \omega^2
 \end{aligned}$$

Quantity	Value	Unit
Outer radius	0.005	m
Inner radius	0.004	m
Length	0.1	m
Density	8000	kg/m ³
Metal area	2.82743338823e-05	m ²
Volume	2.82743338823e-06	m ³
Mass	0.0226194671058	kg
Spin speed	50000	rpm

Quantity	Value	Unit
Angular speed	5235.98775598	rad/s
Spin period	0.0012	s
Axial inertia	4.6369907567e-07	kg m ²
Angular momentum	0.00242792268267	kg m ² /s
Stored energy	6.35628671946	J

Exact mass-inertia reduction of the defined tube

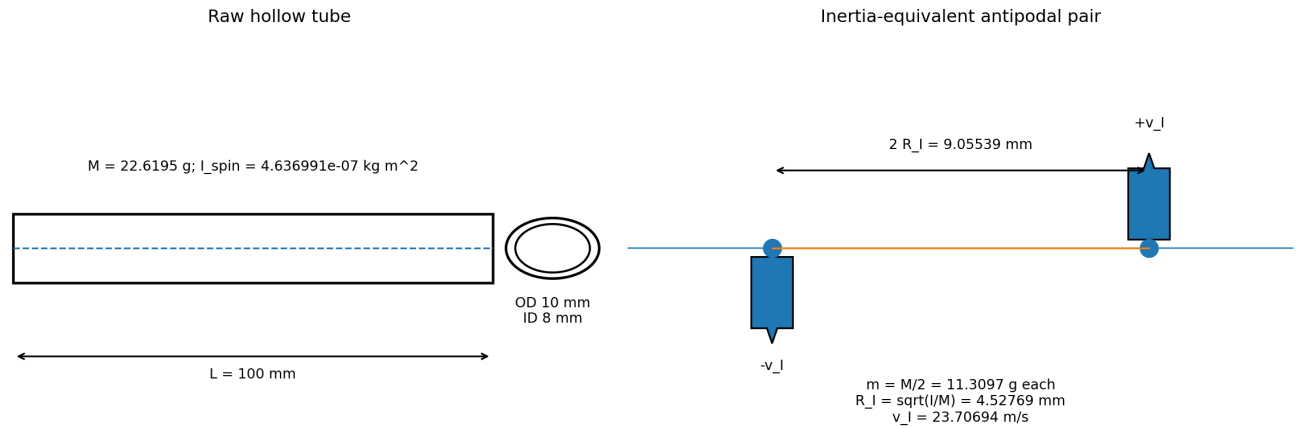


Figure 4. Exact reduction of the hollow tube to two equal point masses at the axial RMS mass radius $R_I = \sqrt{I_{\text{spin}}/M}$. The pair matches the tube mass, axial inertia, angular momentum, and spin energy at the same angular speed.

5. Antipodal reduction

The defined tube can be replaced, for axial mass-inertia accounting, by two equal point masses $M/2$ at positions $\pm R_I$. This pair has the same total mass and axial moment of inertia as the tube. At the same angular speed it therefore has the same spin angular momentum and stored rotational energy.

$$\begin{aligned}
 R_I &= \sqrt{I_{\text{spin}} / M} \\
 m_+ &= m_- = M/2 \\
 I_{\text{pair}} &= 2 (M/2) R_I^2 = I_{\text{spin}} \\
 v_I &= R_I \omega
 \end{aligned}$$

Quantity	Value	Unit
Equivalent radius R_I	0.00452769256907	m
Mass per branch	0.0113097335529	kg
Branch tangential speed	23.7069428545	m/s
Mean branch force	0.000847227793958	N
Peak common branch force	0.00169445558792	N

The pair is not identical to the continuous tube at every instant because it occupies only one antipodal phase. The complete tube occupies all circumferential phases simultaneously. The pair therefore contains a phase-dependent ripple around a steady cycle average; circumferential integration cancels the ripple and leaves the steady average.

6. Turning rate as the primitive variable

Let the signed phase of the rotating state be $\phi(t)$. The signed angular velocity vector is $\Omega(t) = d\phi/dt \hat{n}$. The integrated free acceleration is consequently proportional to the time derivative of orientation phase.

$$\begin{aligned}\Omega(t) &= [d \phi(t) / dt] \hat{n} \\ a_K(t) &= (\kappa_F / 2) \Omega(t) \times g \\ |a_K(t)| &= q_g |\Omega(t)| \text{ when } \hat{n} \text{ is optimal}\end{aligned}$$

A static tube and a rotating tube can have the same instantaneous geometric orientation. They do not have the same force. The static tube has $d\phi/dt=0$ and therefore $a_K=0$. The rotating tube has nonzero phase rate and therefore nonzero force. Geometry determines the directional channel; temporal phase progression activates and scales the channel.

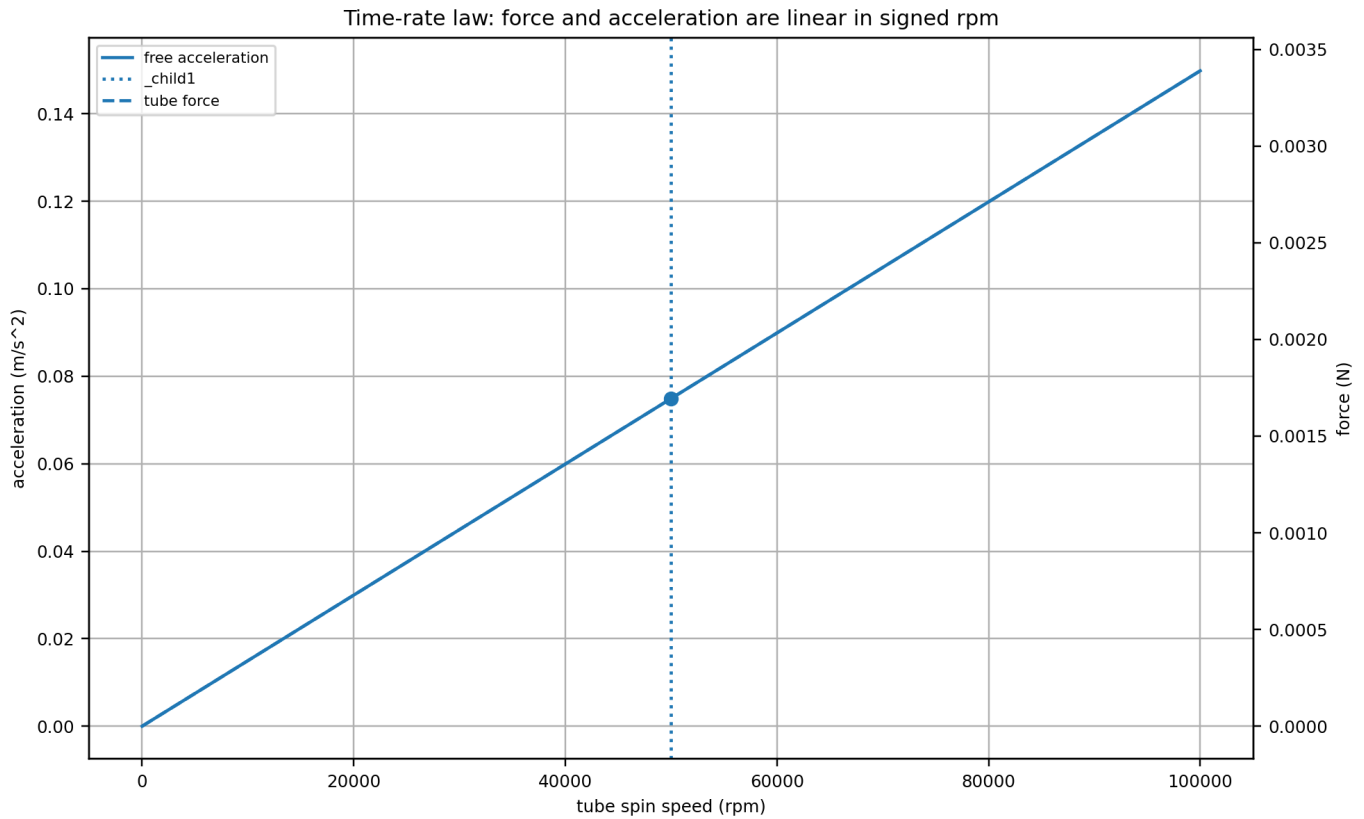


Figure 5. The force and free acceleration are linear in signed tube rpm. The reference point at 50,000 rpm gives 1.69445559 mN and 0.0749113841 m/s².

At optimum orientation, the reference acceleration vector is fixed in ICRS space:

$$\begin{aligned}a_K &= (0.00580335386756, 0.055986960348, 0.0494317389818) \text{ m/s}^2 \\ |a_K| &= 0.074911384074 \text{ m/s}^2 \\ F_K &= (0.000131268771911, 0.00126639520795, 0.00111811959388) \text{ N} \\ |F_K| &= 0.00169445558792 \text{ N}\end{aligned}$$

7. Phase-domain invariant

Multiplying the acceleration law by dt removes the phase rate and reveals the invariant response per signed angular phase.

$$\begin{aligned}d v_K &= a_K dt = (\kappa_F / 2) \hat{n} \times g d \phi \\ d v_K / d \phi &= (\kappa_F / 2) \hat{n} \times g\end{aligned}$$

For a fixed spin axis, the velocity increment depends only on accumulated signed phase. One complete positive revolution has $\Delta \phi = 2 \pi$; one complete negative revolution has $\Delta \phi = -2 \pi$.

$$\Delta v_{\text{per_rev}} = \pi \kappa_F \hat{n} \times g$$

$$|\Delta v_{\text{per_rev}}|_{\text{optimal}} = 8.98936608888\text{e-}05 \text{ m/s}$$

$$|\Delta J_{\text{per_rev}}|_{\text{tube}} = M |\Delta v_{\text{per_rev}}| = 2.0333467055\text{e-}06 \text{ N s}$$

The acceleration is therefore the per-revolution velocity increment multiplied by revolutions per second. At 50,000 rpm the cycle frequency is 833.33333 Hz, and the invariant $8.98936609\text{e-}5 \text{ m/s}$ per revolution is delivered 833.33333 times per second, producing $0.0749113841 \text{ m/s}^2$.

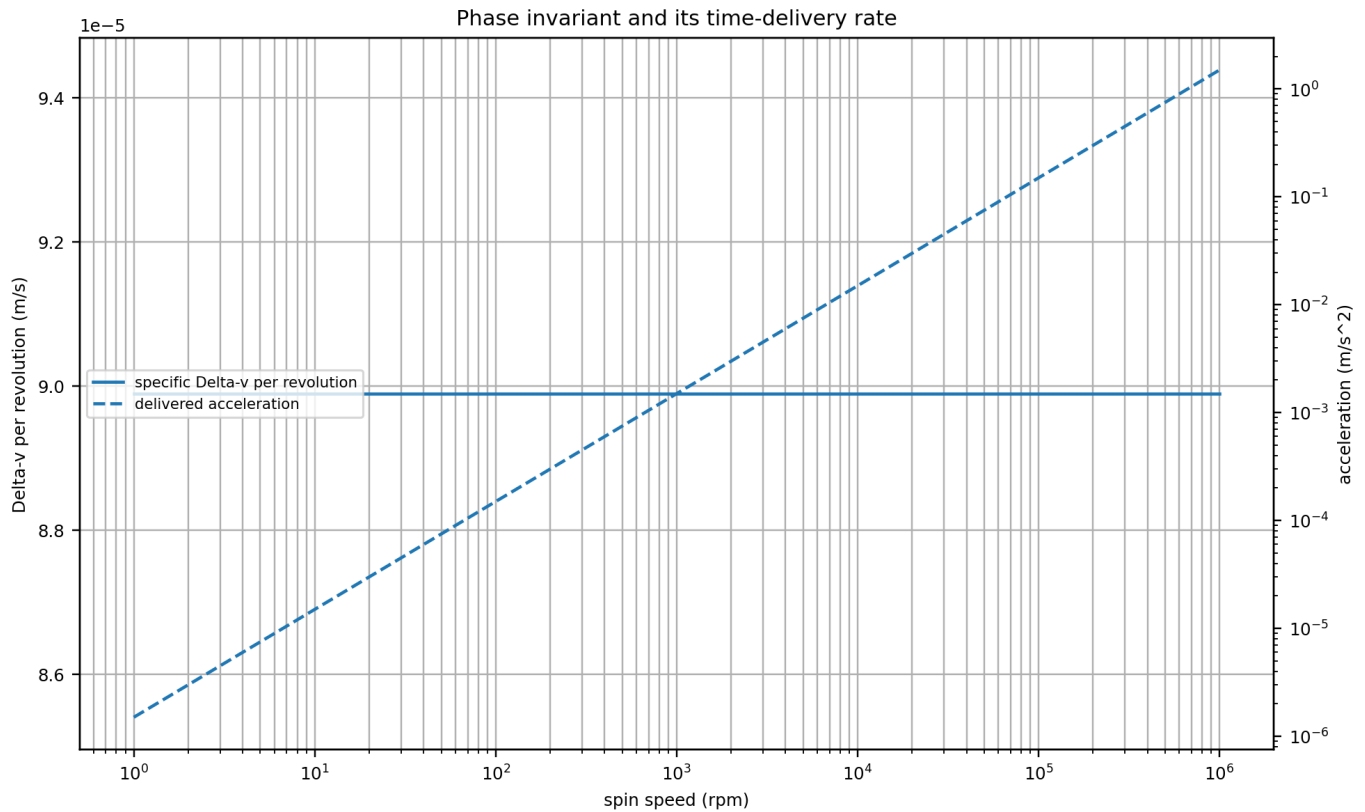


Figure 6. The specific velocity increment per revolution is independent of rpm, while acceleration rises linearly because the invariant phase increment is delivered more frequently.

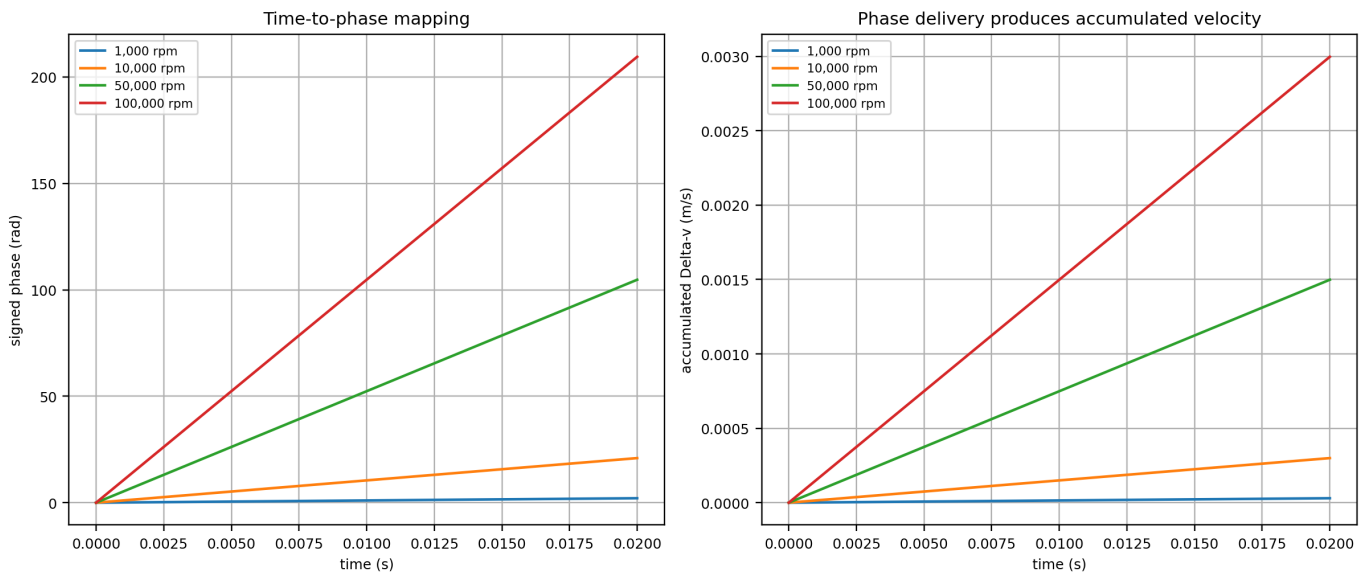


Figure 7. Time-to-phase mapping and the corresponding accumulated velocity for several fixed rotation rates. Faster rotation traverses more signed phase per unit time and therefore delivers the phase invariant more rapidly.

8. Steady operation versus time independence

A uniform rim at constant angular velocity produces a force vector that is constant in inertial space. This is time-invariant operation, but it is not a time-independent constitutive law. The constant vector exists only because the time derivative of phase is held constant.

```
F_K = (kappa_F M / 2) Omega x g
same geometry, different Omega -> different force
Omega = 0 -> F_K = 0
Omega -> -Omega -> F_K -> -F_K
```

Two rotors can occupy the same orientation at the same instant but rotate at different rates. Their forces scale in the same ratio as their angular velocities. A rotor paused at that orientation produces zero force; a rotor traversing it in reverse produces the opposite force. The steady force is therefore a rate-maintained state.

9. Explicit antipodal-pair time response

In the optimal plane, choose e_1 along g and $e_2 = \hat{n} \times e_1$. The instantaneous radial and tangent directions are $u = \cos(\phi)e_1 + \sin(\phi)e_2$ and $t = -\sin(\phi)e_1 + \cos(\phi)e_2$. The common-mode acceleration of the antipodal pair is

```
a_common(phi) = kappa_F omega |g| sin(phi) [cos(phi)e1 + sin(phi)e2]
= (kappa_F omega |g| / 2) [sin(2phi)e1 + (1-cos(2phi))e2]
```

This response is the sum of a constant mean and a rotating ripple.

```
a_mean = (kappa_F omega / 2) n_hat x g
a_ripple = (kappa_F omega |g| / 2) [sin(2phi)e1 - cos(2phi)e2]
ripple angular frequency = 2 omega
```

The ripple and the mean have equal magnitude at the optimal orientation. The instantaneous common-mode magnitude therefore ranges from zero to twice the steady rim value. For the reference tube-equivalent pair, the mean magnitude is 0.0749113841 m/s² and the peak is 0.149822768 m/s².

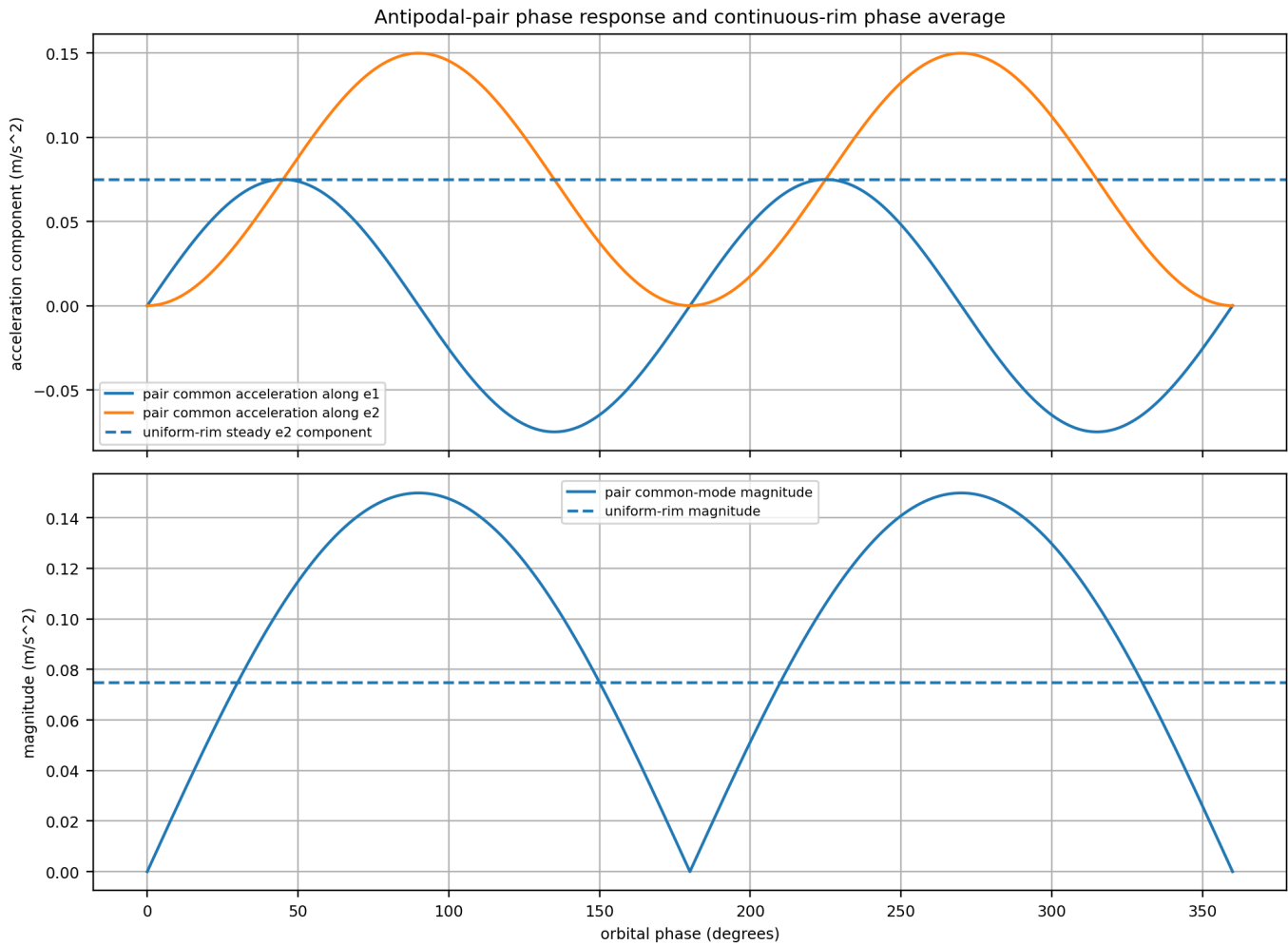


Figure 8. Explicit phase response of one antipodal pair. The continuous rim is the all-phase average: it removes the 2ω ripple and preserves the constant mean vector.

10. Circumferential integration

Every point on a uniform circumference has an antipodal partner. Opposite position and tangent directions both reverse, causing the Earth-velocity cross-term forces to add. The self-motion term is even in angular speed and becomes a differential internal channel that cancels in the net translation.

$$\begin{aligned}
 u(\phi + \pi) &= -u(\phi) \\
 t(\phi + \pi) &= -t(\phi) \\
 [t(\phi + \pi)^T \ g] \ u(\phi + \pi) &= [t(\phi)^T \ g] \ u(\phi) \\
 F_{\text{rim}} = \langle F_{\text{pair}} \rangle_{\text{cycle}} &= (\kappa_F M \omega / 2) \ n_{\text{hat}} \times g
 \end{aligned}$$

The uniform tube wall is therefore not a distinct force mechanism. It is the simultaneous phase completion of the discrete antipodal pair. The pair expresses time-dependent phase structure; the rim expresses its spatially simultaneous average.

11. Time reversal and signed phase

Under time reversal, the signed phase trajectory reverses and $d\phi/dt$ changes sign. The K-Field force is therefore time-odd. Reversing the tube spin reverses the complete force vector by 180 degrees without changing its magnitude.

```

t -> -t
d phi/dt -> -d phi/dt
Omega -> -Omega
a_K -> -a_K

```

A general variable-speed history follows the same rule. During spin-up the force rises in proportion to angular speed. During constant rpm it is constant. During spin-down it decreases. At zero speed it vanishes. During reverse rotation it changes sign.

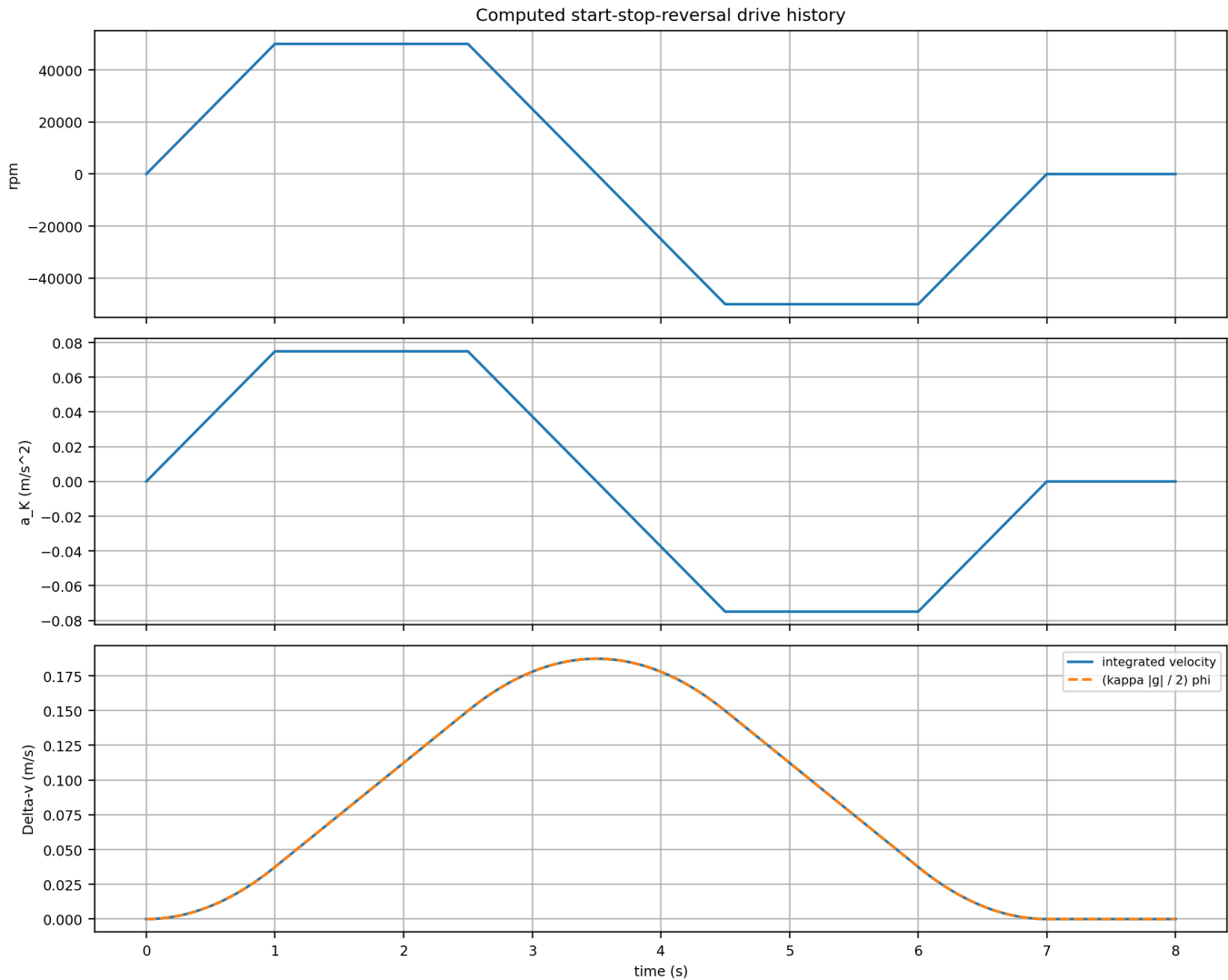


Figure 9. Computed start-stop-reversal history. The acceleration follows signed rpm instantaneously under the linear law. The integrated velocity matches $(\kappa_F |g| / 2)$ times accumulated signed phase.

A forward angular displacement followed by an equal reverse angular displacement has zero net signed phase. Under a fixed axis and field vector, the net K-Field velocity impulse is therefore zero. Unsigned rotation count is insufficient; the response is controlled by signed phase.

12. Large-radius curvature-to-line limit

For a remote-center antipodal pair, the angular turning rate is $\omega = v/R$. The controlled line limit holds finite branch speed v while R increases without bound. Curvature $1/R$, heading change, ordinary centripetal acceleration v^2/R , and the K-Field turning-rate acceleration all tend to zero.

```

R -> infinity
v = constant
omega = v/R -> 0
a_K = (kappa_F / 2) (v/R) n_hat x g -> 0
a_c = v^2/R -> 0

```

The loss of force is not caused by distance alone. It is caused by the vanishing rate of directional change. A mass can retain finite linear speed while its velocity direction changes arbitrarily slowly; the force follows the turning rate, not the translational speed alone.

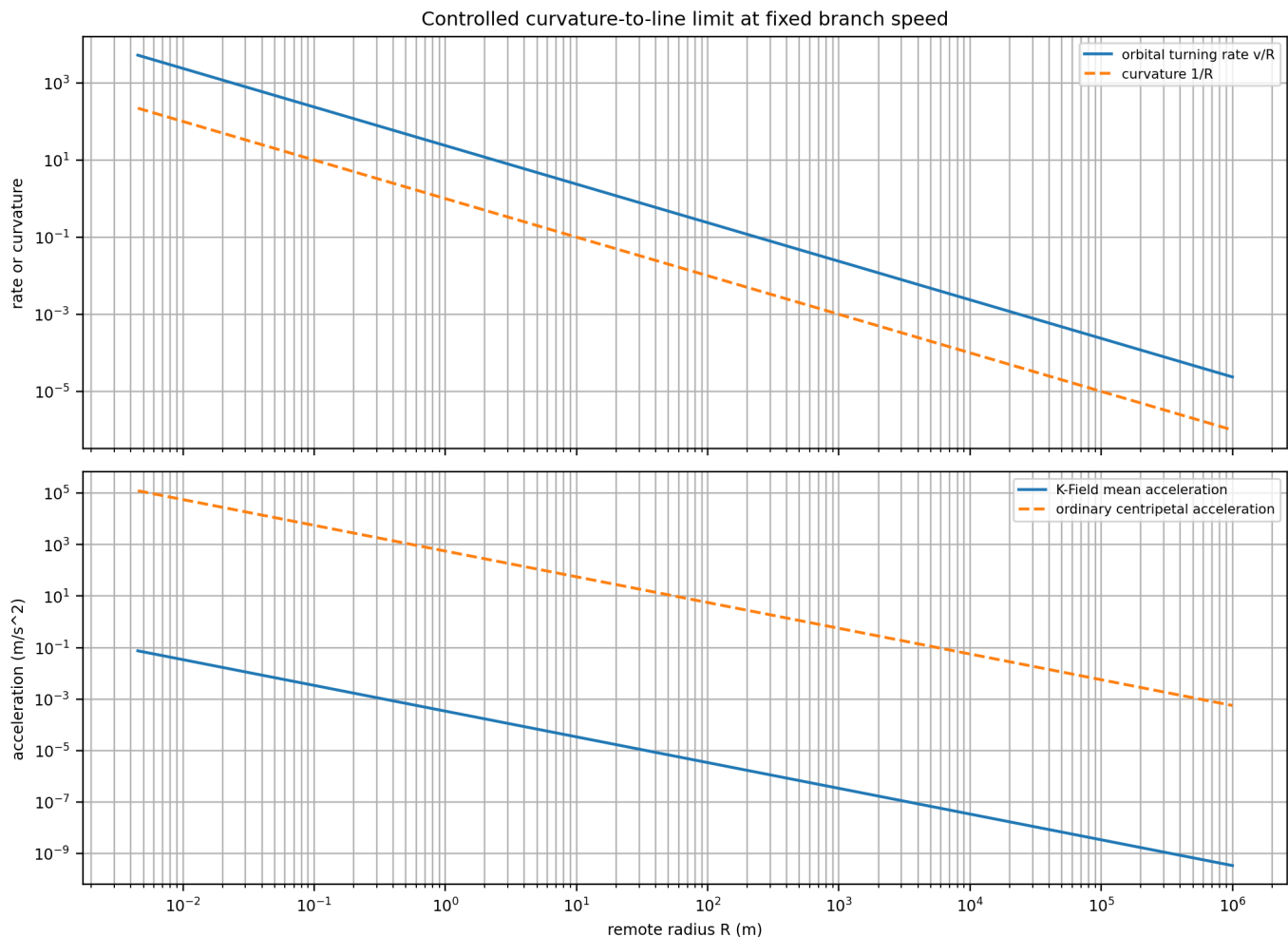


Figure 10. Fixed-speed curvature-to-line limit using the inertia-equivalent branch speed of the defined tube. Both the K-Field mean acceleration and ordinary centripetal acceleration decrease as $1/R$.

This separates two distinct limits. At fixed v , the motion becomes linear and the force vanishes. At fixed ω , the K-Field acceleration remains finite, but $v=R \omega$, ordinary centripetal acceleration, angular momentum, and energy increase without bound; that is not a finite-speed line limit.

13. Accumulated free-space motion

When the tube axis, field vector, and angular speed remain fixed, the integrated tube acceleration is constant in inertial space. Starting from initial velocity v_0 and position x_0 ,

$$\begin{aligned}
 v(t) &= v_0 + a_K t \\
 x(t) &= x_0 + v_0 t + (1/2) a_K t^2
 \end{aligned}$$

Starting from rest at the 50,000 rpm reference state, the scalar speed increment and displacement along f_{hat} are

$$\Delta v(t) = 0.074911384074 \, t \, \text{m/s}$$

$$\Delta x(t) = 0.037455692037 \, t^2 \, \text{m}$$

Elapsed time	Speed increment	Displacement
1 s	0.0749113841 m/s	0.037455692 m
10 s	0.749113841 m/s	3.7455692 m
60 s	4.49468304 m/s	134.840491 m

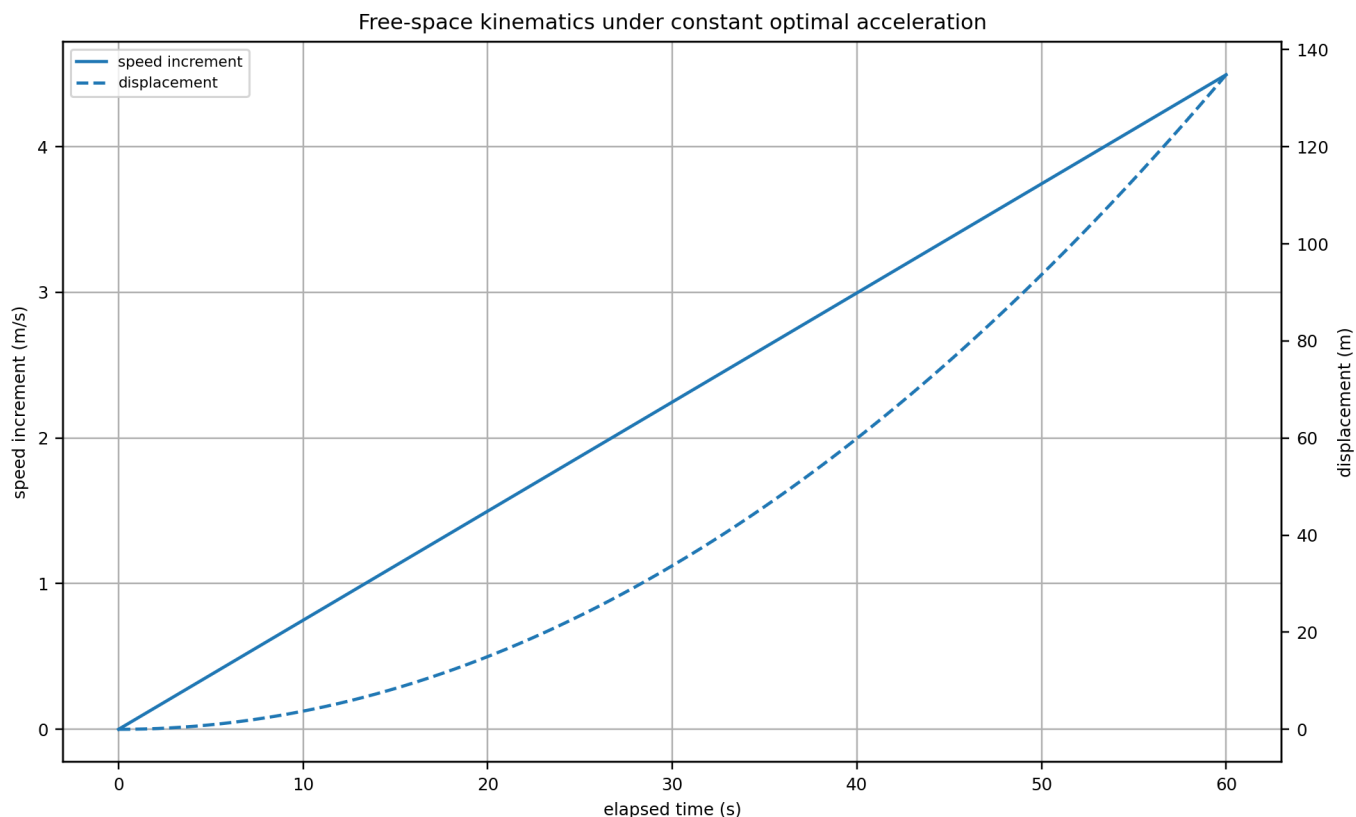


Figure 11. Computed free-space velocity and displacement under the constant optimal acceleration generated by the defined tube at 50,000 rpm.

There are therefore two nested temporal operations. The first derivative $d\phi/dt$ generates acceleration. A subsequent time integral accumulates velocity, and a second integral accumulates displacement.

14. Spin-state and energy relations

The axial spin inertia does not directly multiply the free acceleration law. It determines the angular momentum and energy needed to realize a chosen angular phase rate. At fixed tube geometry,

$$\begin{aligned} a_K &= q_g \, \omega \\ H_{\text{spin}} &= I_{\text{spin}} \, \omega \\ E_{\text{spin}} &= (1/2) \, I_{\text{spin}} \, \omega^2 \\ a_K &= q_g \, H_{\text{spin}} / I_{\text{spin}} \\ a_K &= q_g \, \sqrt{2 \, E_{\text{spin}} / I_{\text{spin}}} \end{aligned}$$

Thus acceleration is linear in angular momentum and proportional to the square root of stored spin energy for fixed geometry. Force is proportional to tube mass, while free acceleration is independent of mass because the mass cancels between F and M .

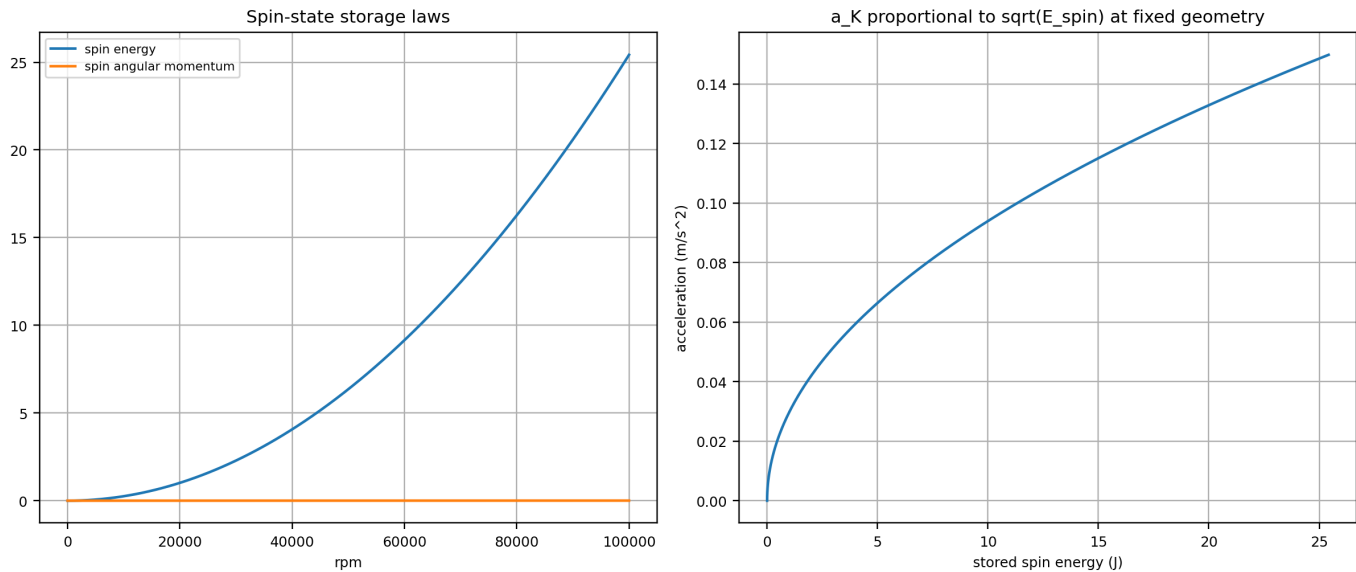


Figure 12. Spin angular momentum grows linearly with rpm, stored energy grows quadratically, and the K-Field acceleration grows as the square root of stored energy at fixed tube geometry.

For the defined tube, the exact reference values are:

Quantity	Value	Unit
I_spin	4.6369907567e-07	kg m^2
H_spin	0.00242792268267	kg m^2/s
E_spin	6.35628671946	J
a_K	0.074911384074	m/s^2
F_K	0.00169445558792	N
F_K/E_spin	0.000266579476776	N/J

15. Dimensional and structural interpretation

For a path with speed v and curvature $K=1/R$, the turning rate is $\omega=v K$. The force law may therefore be written as a curvature-weighted velocity response.

$$\begin{aligned}\omega &= v K \\ a_K &= (\kappa_F / 2) v K \hat{n} \times \mathbf{g}\end{aligned}$$

Neither speed nor curvature alone is sufficient. If $v=0$, no phase is traversed. If $K=0$, the trajectory is straight and its velocity direction does not turn. The active variable is $vK=d\phi/dt$.

For rigid-body rotation, the vorticity is $\zeta=2 \Omega$. The same law can be expressed as

$$a_K = (\kappa_F / 4) \zeta \times \mathbf{g}$$

This form identifies the continuous-rim result as a vorticity or orientation-turning-rate coupling. Opposed linear velocities preserve the projective and momentum geometry of the remote antipodal orbit, but straight translation has zero curvature and zero turning rate.

Dimensional check

Object	Dimension	Role
κ_F	s/m	calibrated response coefficient in the present force law
g	m/s	field-transformed Earth-CMB velocity
ω	1/s	signed angular phase rate
$\kappa_F \omega g$	m/s ²	acceleration
$M \kappa_F \omega g$	N	force

16. Experimental observables

The time-force interpretation produces several direct signatures that separate the response from a static orientation effect. The same apparatus can test all of them without changing tube mass or geometry.

Observable	Predicted signature
Linear rpm law	Measure force while sweeping signed rpm. The slope should be constant and the intercept should pass through zero.
Immediate reversal	Reverse tube spin. The complete force vector should reverse by 180 degrees with unchanged magnitude at equal $ \text{rpm} $.
Zero-speed null	Hold the tube at the same optimal orientation without rotation. The force should vanish.
Impulse per revolution	Integrate acceleration over an integer number of signed revolutions. Delta-v per revolution should be independent of rpm.
Pair ripple	Use two concentrated antipodal masses instead of a uniform tube. The common-mode response should contain a 2 omega ripple around the rim mean.
Rim smoothing	Restore circumferential mass continuity. The 2 omega ripple should cancel while the steady mean remains.
Large-radius fixed-v law	Increase remote radius while reducing omega to hold v fixed. The response should decrease as 1/R.
Signed phase cancellation	Apply equal forward and reverse angular phase. Net velocity impulse should return to zero under a fixed axis and field vector.

A practical measurement must distinguish the predicted lateral channel from ordinary imbalance, bearing reaction, aerodynamic forces, electromagnetic coupling, and gyroscopic transients. The most discriminating controls are signed-rpm reversal, exact zero-speed comparison at the same orientation, and phase-counted impulse measurements.

17. Conclusions

The generated force is not merely associated with a rotating shape. It is generated by the signed time rate of orientation phase. The native geometry fixes an antipodal projective channel; the tube fills every phase of that channel; the Earth-CMB field response fixes the force direction; and angular phase rate determines the delivery rate.

```
Cell3 antipodal geometry
-> rotating antipodal pair
-> all-phase tubular rim
-> phase-domain Delta-v invariant
-> time-domain acceleration
-> accumulated velocity and displacement
```

The complete constitutive statement is

$$a_K(t) = (\kappa_F / 2) [d \phi(t)/dt] \hat{n} \times [G_{\text{phase}} V_E]$$

The geometric invariant is the velocity increment per signed angular phase. Physical force and acceleration are the rate at which that invariant is delivered. Consequently, no angular phase evolution means no force; reversing temporal phase evolution reverses the force; and the force magnitude measures angular phase traversed per unit time.

In the true fixed-speed linear limit, curvature and turning rate vanish together, so the rotational acceleration vanishes. The deepest reduction is therefore a turning-rate or vorticity response, not a translational-speed response.

Appendix A. Matrix and vector ledger

Physical response matrix G_{phase}

row	1	2	3
1	2.976410637800	1.243150779500	-2.080360035400
2	1.243150779500	2.592228663500	-2.317883993700
3	-2.080360035400	-2.317883993700	6.773570913300

Cell3 study-frame basis $B=[a \ b \ c]$

row	a	b	c
1	0.00345107276053829	0.02328032190911276	0.04930856589985942
2	-4.684457135604902e-06	0.02810433453991891	2.287968679773947e-19
3	0.02050872323810781	4.877557189182407e-19	-1.975781252950722e-19

ICRS vectors

Vector	x	y	z	Unit
V_E	-359003.199066	76679.886034	-44701.0601007	m/s
$g=G_{\text{phase}} \ V_E$	-880221.981635	-143911.436562	266335.026949	m/s
c_hat	-0.970770855836	0.207348009111	-0.120874929482	unit
n_opt	-0.315865036407	0.646162683024	-0.694768354089	unit
f_hat	0.0774695854215	0.747375863362	0.65986951907	unit

Appendix B. All 27 Cell3 nodes

L	lambda	chi	kind	study x	study y	study z
(-1, -1, -1)	-3	-1	corner	-0.0380199802848	-0.0140498250414	-0.0102543616191
(-1, -1, 0)	-2	1	edge center	-0.0133656973348	-0.0140498250414	-0.0102543616191
(-1, -1, 1)	-1	-1	corner	0.0112885856151	-0.0140498250414	-0.0102543616191
(-1, 0, -1)	-2	1	edge center	-0.0263798193302	2.3422285678e-06	-0.0102543616191
(-1, 0, 0)	-1	-1	face center	-0.00172553638027	2.3422285678e-06	-0.0102543616191
(-1, 0, 1)	0	1	edge center	0.0229287465697	2.3422285678e-06	-0.0102543616191
(-1, 1, -1)	-1	-1	corner	-0.0147396583756	0.0140545094985	-0.0102543616191
(-1, 1, 0)	0	1	edge center	0.00991462457429	0.0140545094985	-0.0102543616191
(-1, 1, 1)	1	-1	corner	0.0345689075242	0.0140545094985	-0.0102543616191
(0, -1, -1)	-2	1	edge center	-0.0362944439045	-0.01405216727	-1.45088796812e-19
(0, -1, 0)	-1	-1	face center	-0.0116401609546	-0.01405216727	-2.43877859459e-19
(0, -1, 1)	0	1	edge center	0.0130141219954	-0.01405216727	-3.42666922107e-19
(0, 0, -1)	-1	-1	face center	-0.0246542829499	-1.14398433989e-19	9.87890626475e-20
(0, 0, 0)	0	1	body center	0	0	0
(0, 0, 1)	1	-1	face center	0.0246542829499	1.14398433989e-19	-9.87890626475e-20
(0, 1, -1)	0	1	edge center	-0.0130141219954	0.01405216727	3.42666922107e-19
(0, 1, 0)	1	-1	face center	0.0116401609546	0.01405216727	2.43877859459e-19
(0, 1, 1)	2	1	edge center	0.0362944439045	0.01405216727	1.45088796812e-19
(1, -1, -1)	-1	-1	corner	-0.0345689075242	-0.0140545094985	0.0102543616191
(1, -1, 0)	0	1	edge center	-0.00991462457429	-0.0140545094985	0.0102543616191
(1, -1, 1)	1	-1	corner	0.0147396583756	-0.0140545094985	0.0102543616191
(1, 0, -1)	0	1	edge center	-0.0229287465697	-2.3422285678e-06	0.0102543616191
(1, 0, 0)	1	-1	face center	0.00172553638027	-2.3422285678e-06	0.0102543616191
(1, 0, 1)	2	1	edge center	0.0263798193302	-2.3422285678e-06	0.0102543616191
(1, 1, -1)	1	-1	corner	-0.0112885856151	0.0140498250414	0.0102543616191
(1, 1, 0)	2	1	edge center	0.0133656973348	0.0140498250414	0.0102543616191
(1, 1, 1)	3	-1	corner	0.0380199802848	0.0140498250414	0.0102543616191

Appendix C. Formula ledger

Object	Formula
Cell3 node	$p_L = B L / 2$
Antipodal theorem	$S_R \text{ intersection } \Lambda = \{p, -p\}$
Angular phase rate	$\Omega(t) = d\phi/dt \ n_{\hat{}}$
Tube force	$F = (\kappa_F M / 2) \Omega \times g$
Free acceleration	$a = (\kappa_F / 2) \Omega \times g$
Phase invariant	$d v / d\phi = (\kappa_F / 2) n_{\hat{}} \times g$
Per-revolution increment	$\Delta v_{\text{rev}} = \pi \kappa_F n_{\hat{}} \times g$
Optimal magnitude	$a_{\text{max}} = (\kappa_F g / 2) \Omega $
Pair mean	$\langle a_{\text{pair}} \rangle = (\kappa_F \Omega / 2) n_{\hat{}} \times g$
Pair ripple frequency	$\Omega_{\text{ripple}} = 2 \Omega$
Remote-center rate	$\Omega = v/R$
Curvature form	$a = (\kappa_F / 2) v K n_{\hat{}} \times g$
Vorticity form	$a = (\kappa_F / 4) \zeta \times g$
Free velocity	$v(t) = v_0 + \int a \ dt$
Free position	$x(t) = x_0 + \int v \ dt$
Tube mass	$M = \rho \pi (r_o^2 - r_i^2) L$
Spin inertia	$I = (M/2)(r_o^2 + r_i^2)$
Spin momentum	$H = I \ \Omega$
Spin energy	$E = (1/2) I \ \Omega^2$
Equivalent pair radius	$R_I = \sqrt{I/M}$

Appendix D. Source and reproducibility record

Field	Record
Document title	K-Field Time-Force Dependency
Principal Investigator	P. Dwayne Esterline
Organization	Krytronx
Generation epoch	1784909623
Generated UTC	2026-07-24T16:13:43Z
PDF engine	Python ReportLab
Fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier only
Figure generation	Python Matplotlib from numerical ledgers
Output filename	K-Field_time_force_dependency_1784909623.pdf
Machine-readable filename	K-Field_time_force_dependency_1784909623.json
Coordinate frame	ICRS/J2000 Cartesian plus orthonormal Cell3 study frame
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.

Figure ledger

Figure	File
Title mosaic	00_title_mosaic.png
Centered Cell3 and all 27 nodes	01_cell3_all_nodes.png
Cell3 face-plane families	02_cell3_face_planes.png
ICRS vector geometry	03_vector_geometry.png
Tube-to-antipodal-pair reduction	04_tube_pair_reduction.png
Antipodal pair versus rim time response	05_pair_vs_rim_time_response.png
Acceleration and force versus rpm	06_acceleration_force_vs_rpm.png
Phase invariant and time-delivery rate	07_phase_invariant_delivery_rate.png
Start-stop-reversal response	08_start_stop_reversal.png
Free-space kinematics	09_free_space_kinematics.png
Large-radius time limit	10_large_radius_time_limit.png
Spin storage and acceleration laws	11_spin_storage_and_acceleration.png
Phase-time map	12_phase_time_map.png

End of archival reference.